DETAILED CASE STUDY SCENARIO

Spam Email Classification Using Decision Trees and XGBoost

Case Scenario

A large university uses an email system to communicate with students, faculty, staff, and administrators. Every day, the system receives a large number of emails. These emails include genuine academic announcements, examination information, assignment notifications, administrative messages, newsletters, and personal communication. At the same time, the mailboxes also receive unwanted spam messages.

Some spam emails appear to be ordinary messages at first glance. They may contain subjects such as account verification alerts, urgent security notifications, promotional offers, prize announcements, or requests to click a link. Because these messages can look similar to legitimate emails, manually checking every incoming message is difficult and time-consuming.

The university therefore wants an automated system that can examine the content of an email and classify it as either **Spam** or **Legitimate**. The system should learn from previously labeled emails and identify language patterns that commonly occur in unwanted messages.

Detailed Problem Situation

Consider a student who receives an email with the subject **“Urgent: Your university account requires verification”**. The message claims that the student's account will be suspended unless the student immediately clicks a link and confirms account information.

The email contains several characteristics that may also occur in spam messages: urgent language, requests for immediate action, suspicious links, and unusual wording. However, the system cannot simply classify an email based on one word such as “urgent,” because legitimate university emails may also contain urgent notifications.

To solve this problem, the university can use a machine-learning model trained on a large collection of previously labeled emails. The model learns combinations of words and phrases from the email subject and body. When a new email arrives, the trained model uses these learned patterns to predict whether the email is spam or legitimate.

Case Study Approach

The project uses the CEAS-08 email corpus as the source of labeled email examples. The relevant textual information from each email is prepared for machine learning. The subject and body are combined so that the classifier can consider both the title of the message and its complete textual content.

Before classification, the email text is cleaned so that unnecessary formatting and noise do not interfere with learning. The cleaned text is then represented numerically using TF-IDF. This

representation allows the machine-learning algorithms to work with the words and phrases contained in the emails.

Two different classification approaches are considered. A Decision Tree provides an interpretable tree-based approach, while XGBoost uses gradient boosting with multiple decision trees. Both approaches receive the same prepared text representation so that their classification performance can be compared under the same conditions.

Example Email Decisions

Incoming Email	Possible Classification
“Meeting scheduled for tomorrow at 10 AM. Please find the agenda attached.”	Legitimate
“Congratulations! You have won a prize. Click here immediately to claim it.”	Spam
“Your examination timetable has been updated. Please check the university portal.”	Legitimate
“Urgent! Your account will be closed today. Verify your password using this link.”	Spam

Expected System Behavior

When a new email enters the university's mail system, its subject and body are processed in the same way as the training emails. The resulting text representation is passed to the trained classifier. The classifier produces one of two classes: Spam or Legitimate.

If the system predicts Spam, the email can be directed to a spam folder or flagged for additional review. If the system predicts Legitimate, the email can continue to the user's normal inbox. In a practical deployment, the prediction can support the email filtering process rather than replacing all human review.

Why This Case Matters

The case demonstrates a practical application of machine learning and natural language processing. Instead of manually creating rules for every possible spam message, the system learns patterns from historical examples. This is useful because spam messages can change their wording over time.

The comparison between Decision Tree and XGBoost also demonstrates how different machine-learning algorithms can be applied to the same classification problem. Their performance can be assessed using measures such as accuracy, precision, recall, and F1 score, while 5-fold stratified cross-validation can be used to examine consistency across different data splits.

Case Study Conclusion

In this case, a university is facing the practical problem of receiving a large volume of unwanted email. A machine-learning based spam classifier provides an automated approach for separating

unwanted messages from legitimate communication. The system uses email text as the main source of information, converts that text into numerical features, and applies Decision Tree and XGBoost classifiers.

The case study shows how the proposed project can be used in a real-world email environment to support automatic spam detection and reduce the effort required for manual email filtering.